

CSD415 - Project Phase I - Abstract

Submitted by Group 12
Anandhakrishnan S (CEC23CS033)
Aravind A Kamath (CEC23CS041)
Arjun Manoj (CEC23CS043)
Sangeeth Krishnan S (CEC23CS105)

AI-Powered Fake News & Misinformation Detection System using Explainable AI (XAI)

Abstract

The rapid proliferation of fake news and misinformation across social media and digital platforms has emerged as a critical societal challenge. Fabricated news heavily influences democratic elections, spreads medical myths, and has the potential to incite communal violence. There is an urgent need for accessible tools that enable the general public to verify the authenticity of news articles in real time. This project proposes an AI-powered web application designed to automatically classify news headlines and articles as "Real" or "Fake".

The core classification engine utilizes a robust hybrid Deep Learning architecture, integrating FastText word embeddings for capturing deep semantic meanings, Convolutional Neural Networks (CNN) for extracting local contextual features, and Long Short-Term Memory (LSTM) networks for identifying sequential dependencies in the text. Furthermore, to address the black-box nature of traditional deep learning models and build user trust, the proposed system incorporates Explainable AI (XAI) using LIME (Local Interpretable Model-agnostic Explanations). This allows the system to provide an explainability score, visually highlighting the specific words or phrases that triggered the classification. By delivering high-accuracy predictions alongside transparent reasoning, this project aims to provide a reliable, real-time misinformation verification tool for everyday users.

Base Paper Details

Title: Advancing Fake News Detection: Hybrid Deep Learning With FastText and Explainable AI

Authors: Ehtesham Hashmi, Sule Yildirim Yayilgan, Muhammad Mudassar Yamin, Subhan Ali, Mohamed Abomhara

Publisher: IEEE Access

Year: 2024

DOI: 10.1109/ACCESS.2024.3381038

IEEE Xplore Link: <https://ieeexplore.ieee.org/document/10477989>

Methodology

The system operates through a sequential pipeline connecting the user interface with the deep learning model:

1. **Input Stage:** The user provides text input (news article or headline) or the body (optional) via the web frontend.
2. **Text Preprocessing & Embedding:** The text is tokenized and processed through FastText to generate rich, semantic word embeddings.
3. **Feature Extraction (CNN):** The embeddings are passed through a Convolutional Neural Network layer to capture localized textual features (e.g., n-grams, key phrases).
4. **Sequence Learning (LSTM):** The CNN output is fed into an LSTM layer to retain contextual sequence patterns over the length of the text.

5. Classification: A Softmax layer classifies the text into a binary outcome (Real or Fake) with a confidence percentage.
6. Explainability (XAI): LIME interprets the model's decision and maps feature importance back to the original text.
7. Output Stage: The frontend displays the classification label alongside the highlighted suspicious words justifying the outcome.

Dataset

Primary dataset to be used is the WELFake dataset (publicly available via Kaggle) - contains over 72,000 merged text records of real and fake news articles, providing a highly robust and diverse corpus for training the NLP model. Secondary evaluation may utilize FakeNewsNet or FakeNewsPrediction or any other dataset which is also publicly available.

Technologies To Be Used

1. Machine Learning & Data Processing:
 - Frameworks: PyTorch / Keras (TensorFlow)
 - NLP & Embeddings: FastText (by Meta), NLTK, Spacy
 - Explainable AI: LIME (Local Interpretable Model-agnostic Explanations)
 - Training Environment: Google Colab (T4 GPU) / Jupyter Notebook
2. Backend (API & Model Serving):
 - Framework: Python FastAPI
 - Function: Serves the serialized deep learning model (e.g., .h5 or .pt format) and processes real-time API requests from the client.
3. Frontend (User Interface):
 - Framework: React.js / Next.js
 - Function: Provides a responsive dashboard where users can paste news text, trigger the backend analysis, and view the highlighted LIME explanations and credibility scores.